

Towards Synthetic Generation of Daily Routines and Anomaly Detection with Wearable Sensors for Dementia Care

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Abstract: This paper presents an approach to generating synthetic daily routines and systematically injecting behavioural anomalies to support AI-based monitoring in dementia care. Given the ethical and practical challenges of acquiring labelled data from individuals with dementia, a probabilistic routine generator was developed grounded in observational literature, simulating realistic activity patterns using time-of-day segmentation, activity-specific probability distributions, typical duration ranges, and individualized variability in timing and sequencing, alongside controlled context-sensitive deviations. Using this dataset, we trained a Gated Recurrent Unit (GRU)-based sequential anomaly detection model that achieved high classification performance, including an F1-score of 0.959 for anomalous activity steps. The results demonstrate the feasibility and value of synthetic data in developing personalized monitoring systems for early detection of behavioural changes. This approach enables reproducible model benchmarking and lays the groundwork for real-world deployment in wearable-based dementia care systems.

1 INTRODUCTION

There are more than 10 million new cases of dementia each year worldwide, which implies a new case every 3.2 seconds (Alzheimer's Disease International, 2025). Among the most significant challenges in dementia care is gradual loss of autonomy and disruption of daily routines, which can severely impact both patient well-being and caregiver burden.

Continuous, non-invasive monitoring of physical and behavioural parameters through wearable devices play a vital role in detecting subtle changes in activity patterns that may otherwise go unnoticed in traditional clinical assessments. By providing real-time data on mobility, sleep, and daily activities, these technologies can support early identification of health deterioration, facilitate timely interventions, and im-

prove long-term care outcomes (Freitas et al., 2025). Wearables have also been shown to enhance patient autonomy and reduce caregiver burden by integrating activity recognition and behavioural monitoring into everyday life (Rocha et al., 2024; Carvalho et al., 2025). Their portability, unobtrusiveness, and ability to capture longitudinal data make them adequate for use in dementia care, where continuous assessment of routine and behaviour is critical for effective monitoring.

Despite the promise of wearable technology, developing AI-based systems capable of detecting deviations from normal behaviour remains a challenge. Existing activity recognition models perform well when classifying well-defined activities, but identifying subtle anomalies, such as missed meals, increased sedentary behaviour, or

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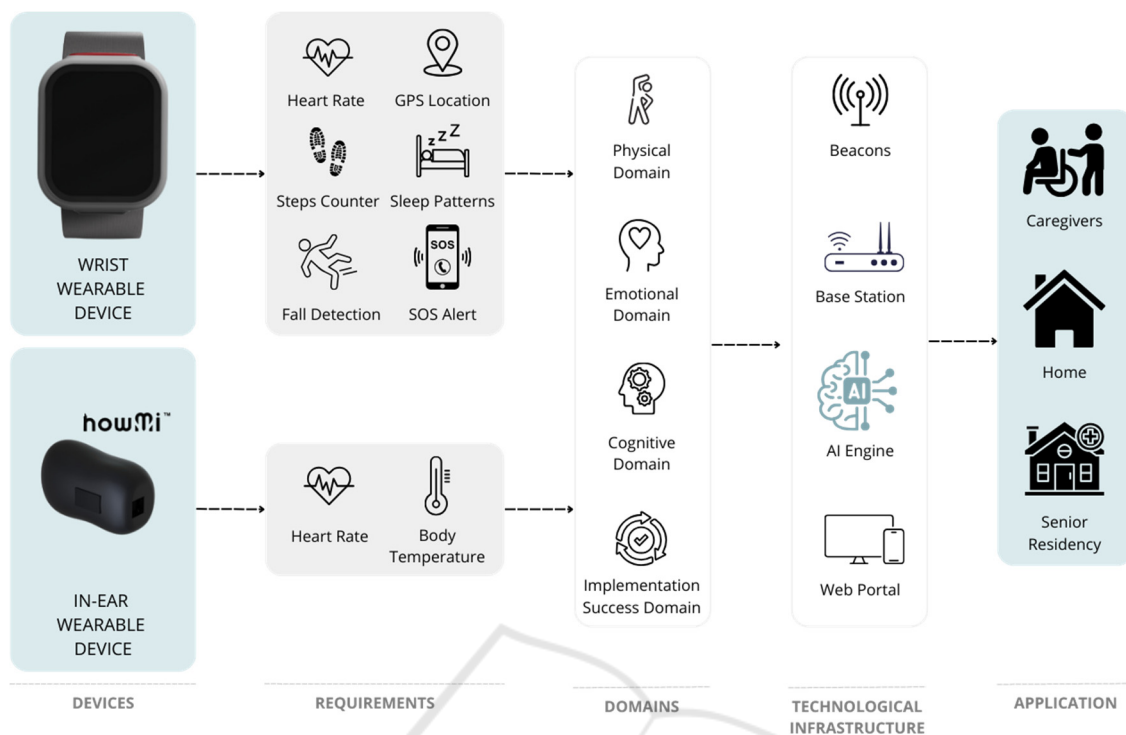


Figure 1: Overview of the wearable-based monitoring system for dementia care.

disorganized routines, requires models to capture long-term temporal patterns and individualized baseline behaviour (Chalapathy and Chawla, 2019). In addition, acquiring sufficiently large and diverse labelled datasets for anomaly detection is particularly difficult in dementia care contexts, due to privacy concerns, ethical constraints, and the rarity and unpredictability of anomalous events (Smuck et al., 2021).

Synthetic data generation comes as a strategy to address these challenges. By simulating realistic daily routines and controlled anomalies, synthetic datasets enable the development and evaluation of anomaly detection models without the need for extensive real-world data collection (Goncalves et al., 2020; Liu et al., 2025).

Building on previous work on wearable-based monitoring for dementia, this study explores the use of synthetic routine generation and AI-based anomaly detection models to support personalised care. The work is part of a broader project aiming to develop intelligent wearable systems (Costa et al., 2023; Arantes et al., 2025) capable of continuously monitoring daily routines and detecting behavioural changes in dementia care contexts. As illustrated in Figure 1, the system combines wearable devices, such as an in-ear sensor and a wrist-worn tracker, with a network of beacons and a base station to

acquire real-time physiological and behavioural data, including heart rate, body temperature, GPS location, and activity patterns.

This paper is organized in 5 sections. Section 2 describes the related work on activity recognition using wearable devices, anomaly detection in behavioural sequences, and synthetic data generation in healthcare. It is followed by a detailed methodology in Section 3 outlining the generation of synthetic routines, simulation of anomalies, and the Gated Recurrent GRU-based detection model. The results are shown in Section 4 presenting performance metrics and analysis. The paper concludes in Section 5 by summarizing contributions and future directions.

2 RELATED WORK

This section reviews relevant literature on three key areas: activity recognition with wearable devices, anomaly detection in behavioural data, and synthetic data generation for healthcare applications.

Wearable devices have increasingly been used to support activity recognition and behavioural monitoring in healthcare. By integrating sensors such as accelerometers, gyroscopes, and physiological monitors, these systems can provide continuous

insights into daily activities, mobility, and sleep patterns. Recent reviews highlight that activity recognition models based on machine learning and deep learning achieve promising results in identifying common behaviours, namely motion, sleep, meals and hygiene related activities (Carvalho et al., 2025). Such capabilities are critical in dementia care, where changes in routine may indicate deterioration of cognitive function or increased health risks. However, most published studies (Ray et al., 2025) rely on datasets from healthy adults, with limited availability of data collected from older adults with dementia, posing challenges for real world deployment.

Beyond recognizing specific activities, detecting deviations from established daily routines is key to enabling proactive interventions in dementia care. Anomaly detection in wearable sensor data involves recognizing patterns that deviate from an individual's typical behaviour, which can be subtle, context dependent, and highly variable (Sunny et al., 2022). According to Sunny et al., anomalies can be categorized as point, contextual, or collective anomalies, each requiring different detection strategies. In wearable data, anomaly detection is typically framed as an unsupervised or semi-supervised learning problem due to the scarcity of labelled anomalous examples. Techniques such as K-Nearest Neighbours (KNN), one-class Support Vector Machine (SVM), autoencoders, Long Short-Term Memory (LSTM)-based autoencoders, sequence-to-sequence models, and transformer-based models have been proposed, with varying degrees of success. Recurrent neural networks, particularly GRUs and LSTMs, are well-suited for modelling long-term temporal dependencies in behavioural data, while attention-based mechanisms can enhance sensitivity to contextual changes.

Detection criteria often rely on reconstruction errors, prediction errors, or probabilistic scores (Blázquez-García et al., 2021). Major challenges remain, including distinguishing true anomalies from noise, managing individual variability, and controlling false positives in real-world deployments (Sunny et al., 2022). In addition to identifying isolated anomalies, an important research direction is the modelling of personalized daily routines and the detection of deviations over extended time frames, which is critical for monitoring progressive changes in dementia care (Shahid et al., 2022).

In dementia care, where real-world data is difficult to obtain and ethically sensitive, synthetic data generation provides a valuable approach for supporting model development and evaluation.

According to Murtaza et al., synthetic data generation methods in healthcare can be broadly categorized as knowledge-driven, data-driven, or hybrid approaches, each offering distinct advantages for addressing data scarcity and privacy concerns (Murtaza et al., 2023). By simulating realistic behavioural sequences and introducing controlled deviations, synthetic datasets enable the training and benchmarking of anomaly detection models under reproducible conditions. This facilitates systematic evaluation of model robustness and generality, contributing to the development of AI-based monitoring systems suited to the complexities of real-world dementia care.

Daily routines play an important role in the well-being and functional stability of people with dementia. Recent observational studies show that, in long-term residential care, the daily structure is strongly organized around communal meals, which act as temporal markers of the day (Gebhard and Frank, 2024). The typical routine identified includes five parts: morning until after breakfast, late morning until lunch, afternoon until dinner, early evening, and night, with large portions of the day, particularly after breakfast and between afternoon coffee and dinner, spent in passive activities or inactivity. Although core activities include hygiene, eating, rest, light physical activity, and social interaction, there is substantial individual variability in the way these routines are experienced. Disruptions in this structure, such as skipped meals or increased inactivity, can indicate cognitive or functional decline. In this study, the synthetic routines were designed to replicate this observed structure and temporal flow, allowing controlled simulation of typical patterns and deviations for model development.

3 METHODOLOGY

This section presents the methodological approach used to generate synthetic daily routines, simulate deviations, and evaluate anomaly detection models. It includes the reasoning for the routine structure construction, anomaly simulation strategies, and the architecture of the AI models used in this study.

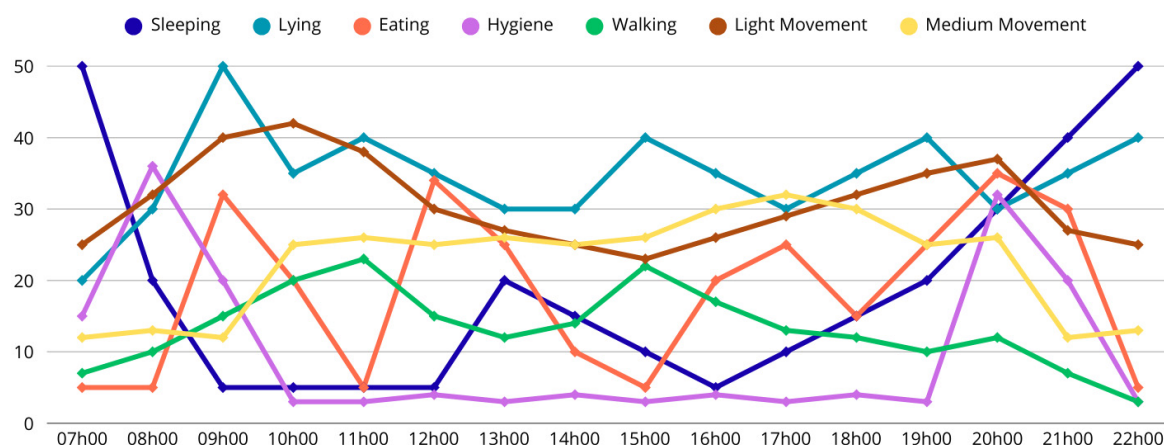


Figure 2: Everyday activities during the course of the day. The percentages (y-axis) refer to the proportion of observed activities at the respective time point.

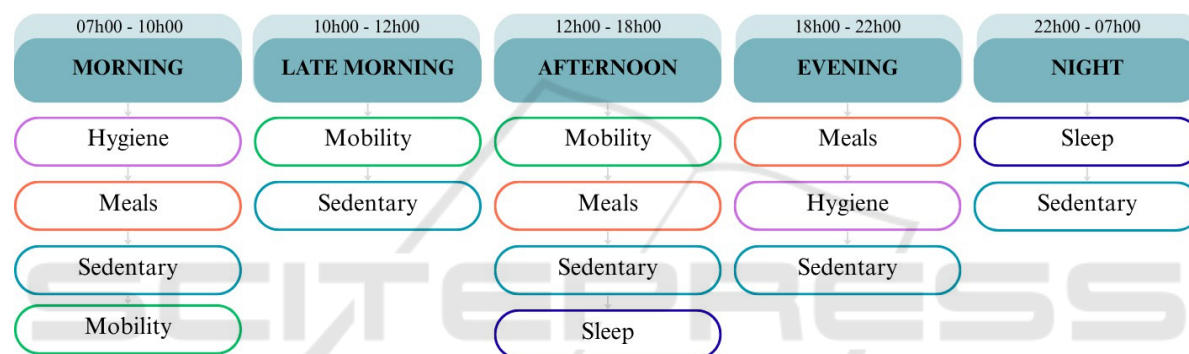


Figure 3: Structure of the daily routine: typical distribution of core activity types across each time-of-day block.

3.1 Synthetic Routine Generation

Synthetic daily routines were generated by dividing each day into 280 time intervals of 5 minutes each, capturing a full 24-hour cycle. This structure was chosen to align with the expected operating frequency of the wearable device, which is designed to communicate updates to detected activity every five minutes. The distribution of activity categories throughout the day is illustrated in Figure 2, which shows the relative proportions of routine activities observed throughout the day. To construct realistic routines, the day was segmented into five broader time blocks: morning (07:00–10:00), late morning (10:00–12:00), afternoon (12:00–18:00), evening (18:00–22:00), and night (22:00–07:00), each characterized by a distinct distribution of activities informed by observational literature on dementia care (Gebhard and Frank, 2024). Within each block, probabilistic rules governed the assignment of activity types such as meals, hygiene, rest, physical activity, and social interaction. Figure 3 illustrates the

typical daily flow used to generate these routines, showing how core activity types are distributed across time-of-day segments to reflect realistic patterns observed in long-term care contexts. Each template was associated with a typical duration range, for example, 10 to 30 minutes for hygiene, 15 to 45 minutes for meals, Figure 4, and random variability was introduced to reflect the flexibility of the routine between individuals. This included small shifts in timing, changes in activity order, and variation in duration, allowing the generation of unique but behaviourally plausible daily schedules.

The types of routines synthesized included:

1. Hygiene: morning and evening routines such as bathing, teeth and hand washing;
2. Meals: breakfast, lunch, dinner, and snacks;
3. Sleep and rest: nighttime sleep and occasional naps;
4. Mobility: low- to moderate-intensity movement such as walking or domestic activity;
5. Sedentary behaviour: laying or sitting without sleeping.

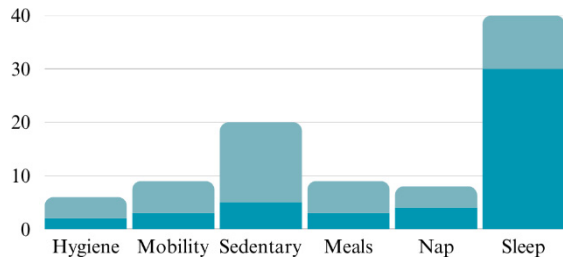


Figure 4: Duration ranges (in number of time steps) of daily activity templates used in the synthetic routine generator. Dark blue bars represent the minimum number of time steps (5-minute intervals), while light blue bars indicate the additional time steps up to the maximum duration.

This approach allows for the creation of synthetic datasets that capture the temporal structure and behavioural diversity of real-world dementia care routines, supporting the training and validation of sequential AI models for anomaly detection.

3.2 Anomaly Simulation

To evaluate the ability of AI models to detect behavioural deviations in daily routines, synthetic anomalies were systematically injected into the generated dataset. The process builds upon the routine generation strategy described earlier and introduces a controlled fraction of anomalous events per day.

Based on the probabilistic routine generation strategy described in the previous section, each day was modelled as a sequence of 280 time steps, representing 5-minute intervals over a 24-hour period. Activities were assigned to each time block using context appropriate templates informed by empirical studies in dementia care.

To simulate anomalies, a specified percentage of the steps in a day, for example 15%, were randomly selected and assigned unexpected activities that do not belong to the distribution of their respective time blocks. For example, inserting “Medium movement” during the night or omitting meals in the afternoon period. These deviations reflect atypical behaviours such as nighttime wandering, missed meals, or activity fragmentation, behaviours often observed in individuals with cognitive impairment.

Each step in the dataset was labelled as either normal (0) or anomalous (1), providing a clear ground truth for evaluation. Two versions of the dataset were created: one with 1000 normal days and another with 10,000 normal plus 10,000 anomalous days (with 15% anomalies per day), enabling both supervised and unsupervised training approaches.

This strategy allows for the reproducible evaluation of anomaly detection models in realistic yet diverse conditions, where deviations are subtle, temporally dependent, and context sensitive.

To illustrate the anomaly injection process, Figure 5 presents a visual comparison between a normal daily routine and one with injected anomalies. The top timeline depicts a typical sequence of activities distributed across the five daily time blocks, including expected behaviours such as sleep during the night and meals at midday. The second timeline shows an altered version of the same day, where certain activities are replaced or shifted out of their usual context, such as unexpected sleep during the morning or missing meals in the afternoon. The bottom timeline displays the corresponding binary labels assigned to each time step (0 for normal, 1 for anomalous), providing the ground truth used for model training and evaluation. This visual

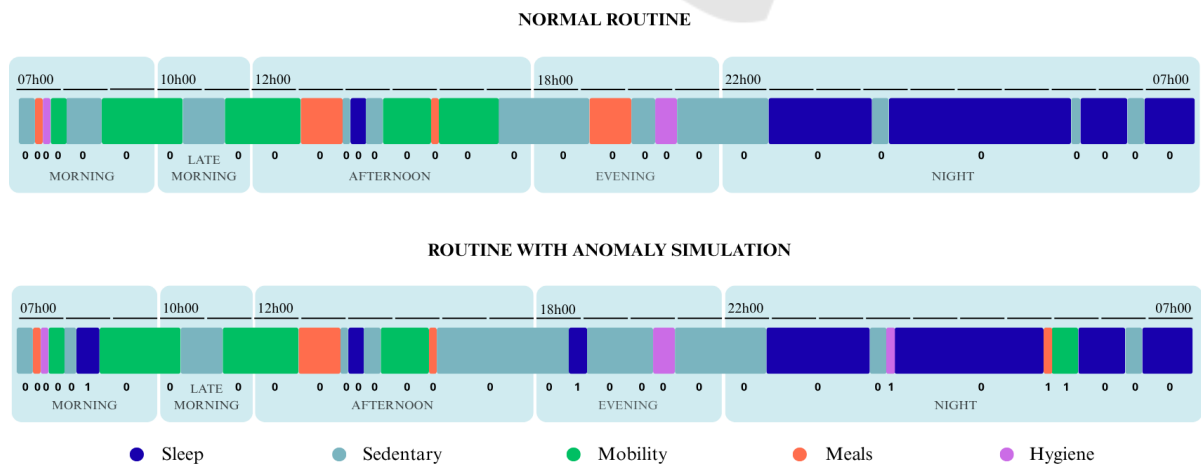


Figure 5: Comparison between a normal routine (top), a routine with injected anomalies (bottom), and the corresponding binary anomaly labels.

representation reinforces the context-aware nature of anomaly simulation, where deviations are not arbitrary but tied to the expected temporal structure of daily routines.

3.3 Anomaly Detection Model

A bidirectional GRU-based sequence model was implemented to learn temporal dependencies in daily activity patterns and identify context-sensitive behavioural deviations. Each daily routine was represented as a sequence of 280 time steps, corresponding to 5-minute activity intervals. The categorical activity labels were first mapped to integer indices and then transformed through a learnable embedding layer (embedding dimension = 8), enabling the model to learn latent relationships between activity types.

To model the temporal dynamics of daily routines, we employed a Bidirectional Gated Recurrent Unit (BiGRU) architecture. The GRU is a recurrent neural network designed to capture long- and short-term dependencies through two gating mechanisms, Figure 6: the update gate and the reset gate. The update gate Z_t controls how much past information should be carried forward, while the reset gate R_t determines how much of the previous hidden state should be forgotten when computing the candidate activation $\tilde{H}_t$.

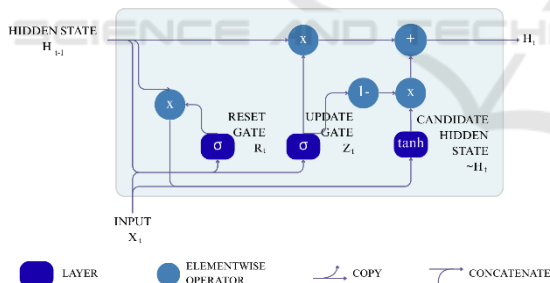


Figure 6: Internal structure of a GRU cell.

This mechanism allows the GRU to selectively retain stable routine information while remaining sensitive to sudden deviations in activity patterns, an essential property for anomaly detection in behavioural sequences.

In the bidirectional configuration, the model processes the sequence in both forward and backward directions. The forward GRU captures dependencies from past to future, while the backward GRU captures temporal relationships from future to past. This produces a context-rich representation that incorporates information from the entire sequence, allowing the model to detect anomalies arising from

irregular ordering, atypical durations, or unexpected transitions between activities.

The embedded sequences were fed into two stacked bidirectional GRU layers, each comprising 128 hidden units per direction. The output of the recurrent layers was passed through a fully connected sigmoid layer that produced an anomaly probability for each time step. A fixed threshold of 0.50 was applied to convert probabilities into binary anomaly labels.

Model training was performed using the AdamW optimizer with a learning rate of 3×10^{-4} and a batch size of 256. A final fully connected layer outputs a probability score for each time step, indicating the likelihood of being anomalous. The model was trained using binary cross-entropy loss, comparing predicted anomaly probabilities to the ground truth labels (0 for normal, 1 for anomalous). Early stopping based on validation loss was used to prevent overfitting, and the performance was evaluated using accuracy, precision, recall, F1-score, and Area Under the Receiver Operating Characteristic (AUROC) curve.

This architecture was chosen for its balance between simplicity, interpretability, and ability to capture sequential structure. It supports both pointwise and contextual anomaly detection by learning not just activity identity, but also its position and timing within the daily routine.

4 RESULTS

The anomaly detection model was trained and evaluated using the synthetic dataset described previously, consisting of 20,000 daily sequences, each comprising 280 time steps of 5-minute activity intervals. Each time step was labelled as either normal or anomalous based on whether it followed the expected behavioural structure for its respective time block. Anomalies were injected into 10,000 of the sequences, with 15% of time steps per day randomly selected for context-inappropriate activities.

The dataset was split into 70% for training, 15% for validation, and 15% for testing. Only normal sequences were used during training and validation, simulating a semi-supervised anomaly detection scenario where the model learns baseline behaviour from healthy routine patterns.

The detection performance was assessed using standard classification metrics, including accuracy, precision, recall, and F1-score. Classification metrics at the step level are summarized in Table 1. The model achieved an overall accuracy of 99.4%. For normal steps, the precision was 0.995 and recall 0.998,

indicating very few false positives. For anomalous steps, the model achieved a precision of 0.978 and recall of 0.941.

Table 1: Performance metrics for normal and anomalous classes.

	Precision	Recall	F1-score
Normal	0.995	0.998	0.997
Anomalous	0.978	0.941	0.959

To assess performance under class imbalance, two additional metrics were computed: the AUROC and the average precision score (PR-AUC). The model achieved an AUROC of 0.999 and a PR-AUC of 0.989, as shown in Figure 7.

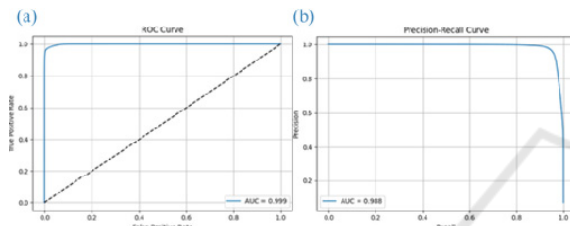


Figure 7: ROC and Precision–Recall curves of the anomaly detection model: (a) AUROC = 0.999; (b) PR-AUC = 0.988.

Figure 8 presents the confusion matrix summarizing the model’s classification performance on the test set.

TRUE LABEL	PREDICTED LABEL	
	normal	anomalous
normal	92.50%	0.16%
anomalous	0.44%	7.05%

Figure 8: Confusion matrix (percentages).

To better understand the model’s behaviour, we analysed the distribution of false positives by activity type. Most false positives occurred during Hygiene activities (665 times), followed by Laying (304), Meals (127), and Sleep (116). Lower misclassification rates were observed in mobility-related activities Medium Movement (59), Light Movement (39), Walking (26), and Standing (1).

5 DISCUSSION

The results show that the model effectively captured

the temporal organization of daily routines and reliably distinguished normal from context-inappropriate activity transitions. The high AUROC and PR-AUC values indicate strong separability between normal and anomalous steps. A decrease in AUROC is expected when anomalies become more subtle or ambiguous, where the overlap between normal and anomalous behaviour increases, reducing class separability.

The choice of simulating 10,000 daily routines was motivated by the need to ensure sufficient diversity of behavioural patterns and stable statistical evaluation, given the combinatorial variability introduced by timing shifts, duration ranges, and activity sequencing.

Analysis of false positives shows that activities with variable temporal boundaries, such as Hygiene, Lying, Meals, and Sleep, are more challenging to classify, particularly due to overlap between sleep, sedentary states, and transitions across time-of-day blocks.

The semi-supervised setup, in which the model learns routine structure exclusively from normal sequences, reflects realistic conditions where anomalous behaviour is scarce and difficult to annotate, while enabling controlled and reproducible evaluation. Limitations include reliance on a bidirectional architecture, which restricts applicability to offline analysis, and the use of synthetic routines that do not fully capture real-world sensor variability.

6 CONCLUSIONS

This study presents a methodology for synthetic generation of daily routines and systematic anomaly injection to support AI-based behavioural monitoring in dementia care. Using knowledge-informed routine templates and probabilistic simulation, we generated a diverse dataset reflecting realistic behavioural patterns and deviations.

A GRU-based anomaly detection model trained on this dataset successfully identified context-sensitive deviations, demonstrating the feasibility of learning routine structure from synthetic data alone and addressing data scarcity and ethical constraints in dementia research.

The proposed approach provides a reproducible and scalable framework for model development and evaluation. However, validation with real-world multimodal wearable data is required, and the current bidirectional architecture must be adapted for real-time deployment. Future work will focus on clinical

validation, refinement of anomaly definitions, and integration into continuous monitoring platforms for personalised dementia care.

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